

Multiple Choice

1. **Answer:** A

Options: A) $(4 \times 4) + (4 \times 5)$; B) $(4+4) \times (4+5)$; C) $(4+4)+(4+5)$; D) $(4 \times 4) \times (4 \times 5)$

Note: Correct option: A - $(4 \times 4) + (4 \times 5)$

2. **Answer:** D

Options: A) $E(X + Y) = 619$, $\text{var}(X + Y) = 7$; B) $E(X + Y) = 619$, $\text{var}(X + Y) = 10$; C) $E(X + Y) = 619$, $\text{var}(X + Y) = 14$; D) There is insufficient information to answer this question.

Note: Correct option: D - There is insufficient information to answer this question.

3. **Answer:** B

Options: A) 382; B) 397; C) 406; D) 447

Note: Correct option: B - 397

4. **Answer:** B

Options: A) 1.8; B) 18; C) 180; D) 1800

Note: Correct option: B - 18

5. **Answer:** D

Options: A) $25 \div 8$; B) $25 - 8$; C) $25 + 8$; D) 25×8

Note: Correct option: D - 25×8

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '-93'.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'for a large sample, regardless of the distribution of random variable X'.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '110 parts'.

Long Form Response

11. **Answer:** We factor the quadratic, getting $(b-5)(2-b) \geq 0$. The expression is equal to 0 when $b = 5$ or 2 . When $b \leq 2$ or $b \geq 5$, the quadratic is negative. When $2 \leq b \leq 5$, the quadratic is non-negative. Therefore, the greatest value of b for which $(b-5)(2-b) \geq 0$ is $b = 5$.

Note: Students should show all steps in their mathematical solution.

12. **Answer:** 1. Explain why this is the correct answer and provide supporting evidence. Show the calculation or reasoning that leads to this conclusion.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key